

Cement

Lacking the extensive quarries of marble available to the Greeks, the Romans had to create their own durable building material. The ancient Egyptians had used gypsum plaster (though this was not particularly strong), the Mesopotamians used bitumen, and some Greek builders, particularly on the coast of Turkey, had developed a form of cement around 200 BCE – but the Romans perfected it.

The Roman invention was distinguished by the addition of lime, which binds sand, water and clay, and by increasing use of finely ground volcanic lava (from Pozzuoli, near Naples) in place of clay. The resulting 'pozzolanic' cement was the strongest mortar in use prior to the British development of Portland cement in the mid-18th century. Roman concrete was based on a hydraulic-setting cement, which set and became adhesive due to a chemical reaction between its dry ingredients and water. By adding fragments of volcanic rubble to this mixture, the Romans were able to build their great feats of engineering.



Concrete dome of the Pantheon, Rome